

Informe de puerta E4 — run_v1.22_r4 (Paper IV v1.22-r4)

Veredicto actual: **PASS** — **reused_verified_external_build**. Evidencia: 20_EVIDENCE/E4/. Auditor: Claude Opus 5.5, misma sesión que las ejecuciones anteriores; no se ejecutó Lean.

E4 — Formal reproduction (r4)

Status: **PASS** — **reused_verified_external_build**, inherited and revalidated by identity. No new build, Lake, Lean or kernel replay was run.

The original external build is run_v1.2_r1, attempt 2, 2026-09-30, 15:09–21:33 UTC.

- **Console** 10_LOGS/E4_main_console.log (ef5b01c7...): modules_planned 607, pass 607, no failures; 607 lines PASS n 607; EXIT 0; all 19 FREEZE_SCOPE targets PASS.
- **records.jsonl** (0f2d8495...): 608 rows, all PASS, exit 0.
- **ReleaseExportCheck**: 224 #check lines, 0 errors.
- **Annex** (57ce1895...): 50/50 PASS; 633 declarations with standard axioms.
- **Objects**: 607 oleans byte-identical to the author's.
- **Axioms**: 461 *records*, of which 314 are #print axioms outputs and the rest repetitions, **not distinct theorems**. The 18 headline declarations depend on {propext, Classical.choice, Quot.sound}. In the 50 public cones there are 0 non-standard axioms, 0 sorryAx and 0 Erdos81 constants.
- E4_main/SUMMARY.json was overwritten by AuditorChecks and is **not used**.

r4 confirmation. 10_SCRIPTS/e4_reuse_check.py (db2eddb8...) produces E4_REUSE_CHECK.json, **identical** to r3 and r2.

Validity. r4 changes only two ES documentary phrases. Lean, the source ZIP, the annex and the run_v1.2_r1 evidence (1097 files) are unchanged (E0 initial and final).

Limits. Same machine and cache; no clean-room build; no kernel replay; the build was produced by the same auditor and session.

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